

JAY SHAH

+1 (817) 449-3968 · shahjay15102002@gmail.com · linkedin/jay-shah · github/jayshah13 · M.S. CS · GPA 3.90 · May 2026

SUMMARY

Entry-level Data Engineer with M.S. Computer Science and hands-on experience building **ETL/ELT pipelines**, designing **data models**, managing **large-scale datasets** from disparate sources, and ensuring **data quality** end-to-end. Proficient in **Python**, **SQL**, **Apache Airflow**, and **AWS**, with exposure to **Spark**, **Kafka**, **Snowflake**, and modern data stack tools. Strong foundation in distributed systems, schema design, and translating business requirements into reliable data infrastructure.

WORK EXPERIENCE

University of Texas, Arlington

Aug 2024 – May 2026

Graduate Assistant — Data Engineering and Analytics

Arlington, TX

- 60+ research projects were generating data from different sources with no unified pipeline – schemas were inconsistent, quality was unvalidated, and there was no automated way to ingest, transform, or monitor data across submissions.
- Built automated **ETL pipelines** in **Python + SQL** with schema validation and data quality checks at every stage; designed data models for consistent storage and retrieval; created **Plotly** dashboards surfacing pipeline health and data quality metrics; documented all transformation logic and data dictionaries.
- 30% fewer data errors**, **25% faster** processing – reliable, well-documented data infrastructure that any team member could maintain, audit, or extend independently.

Tata Consultancy Services

Jan 2024 – May 2024

Data Engineering Intern

Gandhinagar, India

- 500K+ operational records** from multiple source systems had no unified ingestion pipeline – data was siloed, transformations were manual, and a silent schema error had been corrupting downstream reports for weeks without detection.
- Designed and built production **ETL/ELT pipelines** in **Python + SQL**; implemented **data quality checks** and anomaly detection at ingestion; created **dimensional data models** for analytics consumption; deployed pipeline on **AWS (S3, EC2)** with automated scheduling and monitoring.
- Pipeline reliability improved **30%**, delivery **15% faster** – clean, trusted data flowing automatically to downstream consumers with zero manual intervention.

Celebal Technologies

May 2023 – Sept 2023

Data Engineering Intern

Jaipur, India

- 1M+ transaction records** across three business units were stored in legacy formats with no shared schema standard – data couldn't be reliably joined, quality was unknown, and there was no pipeline documentation for the team to maintain.
- Migrated and standardized records into a unified **SQL** database using **Python (Pandas)** transformation scripts; applied **data normalization**, deduplication, and schema enforcement; built **Power BI** dashboards on top of the clean data layer; authored data dictionaries and governance documentation.
- Data accuracy **18% higher**, manual overhead down **25%** – a clean, documented data foundation that three teams could query and build on confidently.

PROJECTS

Live Data Feed System · Python · Apache Kafka · Apache Spark · Airflow · AWS Kinesis · PostgreSQL · Docker

Dec 2025 – Mar 2026

- Business teams needed real-time visibility into operational data but had no streaming infrastructure – all data arrived in batches hours after events occurred, making timely decisions impossible.
- Designed and built an end-to-end **streaming pipeline**: **Apache Kafka** for event ingestion from multiple producers, **Apache Spark Streaming** for real-time transformation and aggregation, **AWS Kinesis** for managed stream processing, **PostgreSQL** as the serving layer, and **Apache Airflow** DAGs orchestrating the full workflow – all **Docker**-containerized for reproducible deployment across environments.
- End-to-end latency under **5 seconds** on **300K+ events/day**, pipeline reliability **99%+ uptime** – business teams had real-time dashboards replacing the previous batch reports that were always hours stale.

Single Source of Truth Reporting System · Snowflake · dbt · Python · SQL · AWS S3 · Airflow · Great Expectations

Apr – Jul 2025

- Analytics teams were writing ad hoc SQL queries against raw tables with no data model, no testing, and no documentation – every analyst had a different version of the same metric and nobody trusted the numbers.
- Architected a **Snowflake** data warehouse with **star schema** dimensional modeling; built a **dbt** transformation layer with staging, intermediate, and mart models; implemented **Great Expectations** data quality tests at every layer; orchestrated full pipeline with **Airflow**; ingested raw data from **AWS S3** via automated Python loaders.
- Single trusted data model serving **12K+ daily queries**, **100% of metrics** defined and tested in dbt – analysts stopped debating numbers and started making decisions from a documented, version-controlled source of truth.

Automated Data Collection and Storage Infrastructure · Python · Apache Spark · Hadoop · AWS S3 · Airflow · Git

Mar – May 2024

- 1M+ records** from multiple disparate sources had no centralized storage, no schema enforcement, and no reproducible transformation pipeline – data scientists couldn't trust or reuse raw data for modeling.
- Built a **data lake architecture** on **AWS S3** with **Delta Lake** for ACID transactions; wrote **Apache Spark** batch jobs for large-scale transformation and aggregation; orchestrated with **Airflow** DAGs; applied **Hadoop**-compatible partitioning strategies; versioned all pipeline code in **Git** with full documentation.
- Pipeline throughput **35% faster**, data reliability **30% higher** – clean, partitioned, versioned data lake that data scientists could reliably pull from for modeling without needing engineering support.

EDUCATION

University of Texas, Arlington

Aug 2024 – May 2026

M.S. Computer Science · GPA: 3.90/4.00 · Distributed Systems · Data Mining · Cloud Computing · Machine Learning

Arlington, TX

Charusat University

Oct 2020 – May 2024

B.S. Computer Engineering · GPA: 3.78/4.00 · Database Systems · Data Structures · Software Engineering

India

SKILLS

Languages: Python (advanced) · SQL (advanced) · Scala (familiar) · Java (familiar) · Bash · R

Batch Processing: Apache Spark · Hadoop · HDFS · MapReduce · PySpark · Hive · data lake design

Stream Processing: Apache Kafka · Apache Flink (familiar) · AWS Kinesis · Spark Streaming · real-time pipelines

Orchestration: Apache Airflow (advanced) · DAG design · scheduling · dependency management · monitoring

Data Warehouses: Snowflake · Google BigQuery · AWS Redshift · Azure Synapse (familiar) · Databricks (familiar)

Transformation & Modeling: dbt · dimensional modeling · star schema · data vault · ELT/ETL design · data normalization

Data Quality: Great Expectations · schema validation · anomaly detection · data dictionaries · lineage tracking

Cloud: AWS (S3, EC2, Glue, Lambda, Kinesis, RDS) · GCP (BigQuery, Dataflow – familiar) · Azure (Data Factory – familiar)

Databases: PostgreSQL · MySQL · MongoDB · DynamoDB · Delta Lake · Apache Iceberg (familiar)

DevOps: Docker · Kubernetes · Git · GitHub Actions · CI/CD · Terraform (familiar) · infrastructure-as-code

Visualization: Power BI · Tableau · Plotly · Matplotlib · Looker (familiar)